DATA ITEM DESCRIPTION

Title: AIRCRAFT STORE LOADS ANALYSIS REPORT

Number: DI-SESS-82233A
AMSC Number: N10399
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20230627
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2023-018

Use/relationship: This report describes the magnitudes and distributions of flight loads and ground loads imposed onto the aircraft store. The data will be used to evaluate the structural design of the store in order to verify the store has met the system performance requirements in the specified operational environment.

This Data Item Description (DID) contains the format, content, and intended use information for the data product resulting from the work task described by the contract SOW.

This DID supersedes DI-SESS-82233.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

1.1. MIL-STD-8591 Airborne Stores, Suspension Equipment and Aircraft-Store Interface (Carriage Phase) is available in https://quicksearch.dla.mil

2. Format. Contractor format acceptable.

3. Content. The Aircraft Store Loads Analysis Report shall contain the following:

3.1. Flight Loads: Describe the aerodynamic data generated, including all data and analyses used to determine the magnitude and distribution of aerodynamic loads. Describe the data in correlation to initial loads estimates proposed using the methodology described in MIL-STD-8591. Describe how the loads estimates were augmented by the subsequent use of Computational Fluid Dynamics (CFD) analysis, wind tunnel test data, and captive carriage Six-Degree-of-Freedom (6DOF) analysis. Describe the correlation of analytical methods and assumptions with CFD, wind tunnel, and flight test data. Describe the following information:

3.1.1. List of weights, speeds, altitudes, attitudes, load factors, variable geometry configurations, useful load distributions, and other pertinent conditions that were investigated with sufficient amplifying information to show that such investigations adequately covered all critical loads.

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3.1.2. The basis for selection of the design load conditions, including discussion of design features affecting loads, parametric analyses, and loads trend data sufficient to provide a foundation for the validity of the selected conditions.

3.1.3. Comparative tabulated values and curves of loads, shears, bending moments, and torsions showing the more critical nature of one condition versus another.

3.1.4. Details of analyses methods with sample calculations, as appropriate.

3.2. Ground Loads: Describe the loads resulting from all static and dynamic ground loading conditions. Describe all loads and structural responses determined by dynamic analyses for takeoff and landing conditions. Describe all other ground loads determined by static load analyses or by other rational methods. Describe how the loads analysis, augmented by captive carriage 6DOF analysis, was used to meet aircraft store ground loads performance requirements. Describe the following in the report:

3.2.1. List each of the specified takeoff, landing impact, and ground handling conditions with sufficient amplifying information to show that such an investigation adequately covers all critically applied external static and dynamic loads as well as airframe response loads.

3.2.2. Results of parametric studies in graphical form to show the variation on externally applied loads and airframe response loads caused by varying critical parameters, such as sinking speeds, airplane weight and weight distribution, attitudes and angular velocities, rough terrain contours, etc.

3.2.3. Time histories of external loads, response loads, and motions for all critical dynamic loading conditions and tabulation of maximum dynamic and static loads.

3.2.4. Time histories of response loads, such as bending moments, shears, and torsions, with tabulation of maximum response loads.

3.2.5. Amplifying information and discussion to substantiate the criticality of conditions in the ground loads analysis.

3.3. Supporting Equations and Documentation: Describe the equations used for determining static and dynamic loads. Describe the following data to support a dynamic loads analysis:

3.3.1 The critical elastic modes of vibration;

3.3.2. The dynamic response characteristics of the store;

3.3.3. Aerodynamic and inertial loads.

End of DI-SESS-82233A

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